Mahdi Kallel

PhD candidate in reinforcement learning

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Profile

- I am a final-year PhD candidate at the University of Würzburg, advised by Prof. Carlo D'Eramo, and expect to graduate in April 2027. I began my PhD developing policy search and policy optimization algorithms, and this work led me to ask how the networks inside these algorithms actually learn. My research now focuses on the learning dynamics of neural networks and on using them to design better training methods. In particular, I work on memory and plasticity in continual learning and on how these dynamics differ between reinforcement and supervised learning.

Publications

The Spectral Anatomy of Plasticity Loss 2026
Uses random matrix theory to show that learned structure persists as plasticity declines, while later parameter updates increasingly concentrate in existing spectral directions.
Mahdi Kallel, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo
openreview.net/forum?id=Kgz0zlJVS6 (AXIOM Workshop at NeurIPS 2026)

Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement 2026
Trains classifiers to refine their predictions through reinforcement learning, improving the balance between accuracy and calibration compared with cross-entropy. The critic estimates remaining improvement and decides when to stop.
Mahdi Kallel, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo
arxiv.org/abs/2604.22110 (Transactions on Machine Learning Research (TMLR), 2026)

PPO+: Enhancing Proximal Policy Optimization 2026
Shows that the critic is a bottleneck in PPO, and that adding off-policy data for critic training improves on-policy gradient estimates and performance.
Mahdi Kallel, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo
rlj.cs.umass.edu/2026/papers/Paper8.html (Reinforcement Learning Journal (RLC 2026))

Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning 2026
Reduces the learning delay of target networks by taking the minimum of online and target value estimates, enabling faster learning while controlling overestimation.
Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo
arxiv.org/abs/2510.02590 (ICLR 2026)

Augmented Bayesian Policy Search 2024
Introduces a Gaussian process prior that guides policy search with the policy gradient, enabling Bayesian optimization to scale to high-dimensional control problems.
Mahdi Kallel, Debabrota Basu, Riad Akrour, Carlo D'Eramo
proceedings.iclr.cc/paper_files/paper/2024/hash/7eca17ef54789b0663cab421f2e9dbf5-Abstract-Conference.html (ICLR 2024)

Cyclicity-Regularized Coordination Graphs 2024
Shows that accounting for cycles when sparsifying coordination graphs improves joint action selection and performance in cooperative multi-agent reinforcement learning.
Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo
rlj.cs.umass.edu/2024/papers/RLJ_RLC_2024_40.pdf (RLC 2024)

Research Experience

INRIA Lille, SCOOL Research Group, Research Intern — France, 2022
- Started research on Bayesian policy search with deterministic policies, leading to Augmented Bayesian Policy Search (ICLR 2024).

EURECOM, Research Intern — Biot, France, 2020
- Compared centralized and decentralized approaches to mobile traffic prediction and load balancing.

Industry Experience

Mcovision (independent), Machine Learning Engineer — Paris, France, 2021
- Built a classifier of echocardiography videos for cardiac amyloidosis and other rare heart diseases with Hôpital Cochin, from data cleaning to training.

SustainEcho, Research Intern — Paris, France, 2021
- Explored few-shot and hierarchical learning methods for classification with little data.

Education

PhD **University of Würzburg**, Reinforcement Learning — Germany, 2023 – 2027
- Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
- Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

MSc **Université Paris-Saclay**, Optimization — France, 2021 – 2022
- Coursework: Convex analysis, dynamic programming, stochastic optimization and approximation.

MSc **ENS Paris-Saclay**, Mathematics, Vision, Learning (MVA) — France, 2020 – 2021
- Coursework: Reinforcement learning, convex optimization, kernel methods, advanced deep learning.

BSc **Télécom Paris / EURECOM**, Data Science — France, 2018 – 2020

Supervision, Talks and Service

Supervision: Johannes Tölle, MSc thesis, University of Würzburg (2026). The thesis received the VDE Bayern Award 2026, and Johannes co-authored two of the papers above.

Talks: On Better Priors for Policy Search. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer: NeurIPS (top reviewer, 2026), ICLR, ICML, RLC

Technical Skills

Programming: Python, JAX, PyTorch, NumPy, Julia, C

Languages: Arabic (native), English (fluent), French (fluent), German (basic)